

LEAST-COLOUR RAISING DYNAMICS ON FINITE TYPE-A CRYSTAL WORDS

ANONYMOUS

ABSTRACT. Fix an alphabet $[k]$ and a word length n . At each epoch, apply the Kashiwara raising operator e_i of the least colour for which it is defined; if none is defined, hold. This deterministic choice orients every type-A word-crystal component toward its unique highest word. We give the exact entrance time of every word, prove the sharp global depth $n(k-1)$ with unique extremizer k^n , and enumerate every depth layer by a normalized principal specialization of a Schur polynomial. Components of shape λ occur with multiplicity f^λ , so the fixed words are counted by involutions whose Robinson–Schensted shape has at most k rows. Independently, we solve the inverse problem for every labelled target: its predecessors are precisely the admissible one-step lowerings f_i , together with a self-predecessor at a highest word. Every fibre has size at most k , and this bound is attained exactly in the stable range $n \geq \binom{k}{2}$; a staircase highest word gives an explicit witness. Classical crystal, tableau, RSK, Schur, and hook formulas, as well as existing crystal pop-stack sorting, receive zero contribution credit. The retained object is only the least-colour scheduler and its targetwise fibre atlas. A bounded owner search leaves the manuscript `OWNER_AMBER / HOLD_EXTERNAL`.

1. THE LITERAL SCHEDULER AND ITS SUBTRACTION BOUNDARY

Let $n, k \geq 1$ and $\mathcal{W}_{n,k} = [k]^n$. We first fix the signature convention, since reversing a word or reversing the tensor convention changes which occurrence an operator edits. For $1 \leq i < k$ and $w = w_1 \cdots w_n$, ignore letters outside $\{i, i+1\}$, write a plus sign for each i and a minus sign for each $i+1$, and repeatedly delete matched pairs $+-$, with ignored letters allowed between the signs. The remaining signs are a string of minuses followed by a string of pluses. If a minus remains, $e_i(w)$ changes the position of the *rightmost* unpaired $i+1$ into i ; otherwise $e_i(w)$ is undefined. If a plus remains, $f_i(w)$ changes the position of the *leftmost* unpaired i into $i+1$; otherwise $f_i(w)$ is undefined. These partial maps are mutual inverses along every i -string.

Define $F = F_{n,k} : \mathcal{W}_{n,k} \rightarrow \mathcal{W}_{n,k}$ by

$$(1) \quad F(w) = \begin{cases} e_{m(w)}(w), & m(w) = \min\{i : e_i(w) \text{ is defined}\} \text{ exists,} \\ w, & \text{all } e_i(w) \text{ are undefined.} \end{cases}$$

Thus the state is re-examined after every individual crystal edge. For example, when $k = 3$,

$$(2) \quad 333 \xrightarrow{2} 332 \xrightarrow{1} 331 \xrightarrow{2} 321 \xrightarrow{1} 311 \xrightarrow{2} 211 \xrightarrow{1} 111,$$

where the superscript is the selected colour.

Crystal bases, Kashiwara operators, tensor-product signatures, and unique highest vertices are established theory [1, 4]. RSK, tableaux, Schur functions, principal specialization, and hook formulas are likewise classical [3, 5, 6]. We use all of that material as background and claim no contribution for it. In particular, neither the crystal edges nor the decomposition of $[k]^n$ is new here. The only retained conjunction is the autonomous least-colour choice in (1), its exact clock, and its complete labelled inverse atlas. A bounded source non-hit is not evidence of novelty, priority, completeness, or freedom to operate; external circulation remains on hold.

Defant and Williams' crystal pop-stack operator sends a vertex in a poppable crystal to the unique source of its connected component after restricting to all descent colours of the starting vertex [2]. It is a noninvertible deterministic crystal map whose forward orbits reach the highest-weight vertex, with a sharp maximum orbit-size theorem. This is the nearest located dynamical surface and receives zero contribution credit. It is not (1): one pop-stack macrostep resolves a whole descent-colour component using the starting descent set, whereas F takes one edge of the least currently available colour and recomputes after every edge. That work does not supply the letter-sum clock, Schur depth layers, or the targetwise f_i predecessor atlas proved below. These distinctions bound the literal object; they do not assert novelty.

Key words and phrases. crystal graph, Kashiwara operator, finite dynamical system, ballot word, Schur specialization, fibre.

2. COMPONENTS, RECURRENT STATES, AND THE EXACT CLOCK

Write

$$(3) \quad E(w) = \sum_{a=1}^n w_a.$$

With our signature convention, the convenient RSK shape is

$$(4) \quad \text{sh}(w) = \text{shape}(\text{RSK}(w_n \cdots w_1)).$$

This reversal is part of the convention, not a dynamical operation. If $\lambda = (\lambda_1, \dots, \lambda_k)$ is a partition padded by zero parts, put

$$(5) \quad b(\lambda) = \sum_{i=1}^k i\lambda_i.$$

Proposition 2.1 (highest words). *The following are equivalent for $h \in \mathcal{W}_{n,k}$:*

- (i) $F(h) = h$;
- (ii) every $e_i(h)$ is undefined;
- (iii) for every prefix u of h and every $1 \leq i < k$, the number of letters i in u is at least the number of letters $i+1$ in u .

Thus the fixed states are exactly the ballot, or Yamanouchi, words. Every crystal component contains exactly one of them. If its shape is λ , that word has content λ and energy $b(\lambda)$.

Proof. The first equivalence is the holding clause in (1). Under the matching rule, an unpaired minus in the i -signature exists exactly when some prefix has seen more letters $i+1$ than letters i . This proves the second equivalence. The remaining assertions are the highest-weight description of a finite type-A word crystal; with (4), its highest weight is its RSK shape. $\square$

Let $\tau(w)$ be the least $t \geq 0$ for which $F^t(w)$ is recurrent.

Theorem 2.2 (unique sink and pointwise clock). *Every recurrent state is fixed. A word w remains in its crystal component and converges to that component's unique highest word $h(w)$. If $\lambda = \text{sh}(w)$, then*

$$(6) \quad \tau(w) = E(w) - b(\lambda).$$

Consequently,

$$(7) \quad \max_{w \in \mathcal{W}_{n,k}} \tau(w) = n(k-1),$$

and k^n is the unique word attaining this maximum.

Proof. Whenever (1) moves, one letter $i+1$ becomes i , so E drops by exactly one. The move is a crystal edge and hence preserves the component. Strict decrease excludes every nontrivial directed cycle, while finiteness and Proposition 2.1 force the orbit to stop at the unique highest word of its component. Its energy is $b(\lambda)$, proving (6).

Since $E(w) \leq nk$ and $b(\lambda) \geq n$, equation (6) gives $\tau(w) \leq n(k-1)$. Equality requires every letter of w to be k and $b(\lambda) = n$. The first condition already forces $w = k^n$; its reverse-RSK shape is (n) and its endpoint is 1^n , so equality does hold. $\square$

The colour schedule affects the route but not the length: any sequence of raising edges ending at the component's highest word has the energy difference in (6). What is special to (1) begins with the actual chosen route and, decisively, with its inverse problem.

3. EXACT COMPONENT AND GLOBAL LAYER POLYNOMIALS

For a cell $x = (r, c)$ of a Young diagram, with rows and columns numbered from one, let $\text{ct}(x) = c - r$ and let $h(x)$ be its hook length. Write $f^\lambda = |\text{SYT}(\lambda)| = n! / \prod_{x \in \lambda} h(x)$.

Theorem 3.1 (Schur-specialized layers). *Every shape- λ crystal component has depth polynomial*

$$\begin{aligned}
 (8) \quad D_{\lambda,k}(q) &= \sum_{w \text{ in one component}} q^{\tau(w)} \\
 &= q^{-b(\lambda)} s_{\lambda}(q, q^2, \dots, q^k) \\
 &= q^{-\sum_{r \geq 1} (r-1)\lambda_r} s_{\lambda}(1, q, \dots, q^{k-1}) \\
 &= \prod_{x \in \lambda} \frac{1 - q^{k+\text{ct}(x)}}{1 - q^{h(x)}}.
 \end{aligned}$$

There are exactly f^{λ} such components. Hence the global depth polynomial is

$$(9) \quad D_{n,k}(q) = \sum_{\substack{\lambda \vdash n \\ \ell(\lambda) \leq k}} f^{\lambda} D_{\lambda,k}(q),$$

and $[q^d]D_{n,k}(q)$ is the exact number of words at depth d .

Proof. Reverse-word RSK identifies words with pairs (P, Q) of common shape λ , where P is semistandard with entries in $[k]$ and Q is standard. With the convention of Section 1, crystal operators change P and preserve Q . Thus Q indexes a component, giving f^{λ} components, and the energy is the sum of the entries of P . The unique highest tableau has every cell in row r filled by r , of weight $b(\lambda)$. Summing (6) over all P gives the first Schur specialization in (8). Homogeneity removes a factor q^n , and the principal-specialization hook-content identity gives the product. Summing over the recording tableaux proves (9). $\square$

At $q = 1$, the component size is the hook-content value

$$(10) \quad D_{\lambda,k}(1) = s_{\lambda}(1^k) = \prod_{x \in \lambda} \frac{k + \text{ct}(x)}{h(x)},$$

and (9) specializes to k^n .

Corollary 3.2 (fixed words and involutions). *The recurrent and fixed census is*

$$(11) \quad |\text{Fix}(F_{n,k})| = \sum_{\substack{\lambda \vdash n \\ \ell(\lambda) \leq k}} f^{\lambda}.$$

It also counts involutions in S_n whose RSK shape has at most k rows, equivalently whose longest decreasing subsequence has length at most k . For $k \geq n$ this is the ordinary involution number I_n , with

$$(12) \quad \sum_{n \geq 0} I_n \frac{z^n}{n!} = \exp(z + z^2/2).$$

Proof. Each component contributes its unique highest word. Under RSK, involutions are the pairs (Q, Q) , one for each standard tableau Q ; Schensted's theorem identifies the number of rows with longest decreasing subsequence length. The unrestricted recurrence $I_n = I_{n-1} + (n-1)I_{n-2}$ gives (12). $\square$

Equations (8)–(12) are classical crystal/RSK enumerations applied to the clock. They provide exact layers but do not by themselves supply an originality claim.

4. EVERY-TARGET INVERSE ATLAS

For an assertion such as “ $e_j(x)$ is absent”, we mean that the partial operator is undefined on that specific labelled word x .

Theorem 4.1 (complete labelled fibres). *For every target $y \in \mathcal{W}_{n,k}$,*

$$(13) \quad F^{-1}(y) = (\{y\} \text{ if } y \text{ is highest, else } \emptyset) \cup \bigcup_{i=1}^{k-1} \left\{ f_i(y) : \begin{array}{l} f_i(y) \text{ is defined,} \\ e_j(f_i(y)) \text{ is undefined for every } j < i \end{array} \right\}.$$

Each displayed lowering contributes at most one distinct source. In particular,

$$(14) \quad |F^{-1}(y)| \leq k$$

for every target, and the fibres sum to k^n .

Proof. Suppose first that $x \neq F(x) = y$, and let i be the colour selected at x . Then $y = e_i(x)$ and the crystal-string inverse property gives $x = f_i(y)$. The scheduler selected i exactly because every $e_j(x)$ with $j < i$ was absent. Thus x appears on the right of (13). The only possible self-predecessors are the holding, hence highest, states.

Conversely, let $x = f_i(y)$ satisfy the displayed absence conditions. Then $e_i(x) = y$, and no lower colour is available at x , so (1) selects i and sends x to y . A candidate $f_i(y)$ changes the content by the simple root of colour i , so candidates from different colours are distinct. There are at most $k - 1$ of them and at most one self-source, proving (14). Summing fibres over a finite self-map counts each source once. $\square$

The bound has an exact stability threshold.

Corollary 4.2 (stable sharpness). *Assume $k \geq 2$. Some target has fibre size k if and only if*

$$(15) \quad n \geq \binom{k}{2}.$$

For $s = n - \binom{k}{2} \geq 0$, an explicit target attaining the bound is the highest word

$$(16) \quad h_{n,k} = 1^{k-1+s} 2^{k-2} 3^{k-3} \cdots (k-1).$$

For example, $F^{-1}(112) = \{112, 212, 113\}$ when $(n, k) = (3, 3)$. For $k = 1$, the sole fibre has size one.

Proof. A fibre of size k must use all $k - 1$ lowering candidates and the self candidate, so its target is highest. If its padded highest weight is $\lambda = (\lambda_1, \dots, \lambda_k)$, then f_i is defined exactly when $\lambda_i > \lambda_{i+1}$. Strict inequalities for every i force

$$n = \sum_i \lambda_i \geq (k-1) + (k-2) + \cdots + 1 = \binom{k}{2}.$$

Conversely, the word in (16) is ballot and has padded content $(k-1+s, k-2, \dots, 1, 0)$, so every $f_i(h_{n,k})$ exists. For $j < i - 1$, applying f_i does not change the j -signature; in the $(i-1)$ -signature it removes one minus and therefore cannot create an unpaired minus. Since the target is highest, every lower e_j remains absent. All $k - 1$ candidates are therefore admitted by (13), along with the self-source. $\square$

Unlike the Schur layer formula, Theorem 4.1 depends on the least-colour scheduler: another order on the same crystal edges changes its admissibility conditions and its labelled fibres.

5. FINITE CONTROL AND EVIDENCE BOUNDARY

The paper-local verifier constructs the literal signatures and the whole functional graph independently of its tableau routines. On all words for $1 \leq k \leq 4$ and $1 \leq n \leq 7$, it checks the pointwise clock, unique highest endpoint, ballot criterion, reverse-RSK shape invariance, sharp and unique deepest word, and every target's exact predecessor set. It separately enumerates semistandard tableaux and compares their depth polynomials with the hook-content principal specialization, builds crystal components and checks the multiplicity f^λ , verifies the hook-length formula by a corner-removal recurrence, and checks the bounded-height involution census through S_8 . Staircase fibres are tested directly through $k = 9$ without enumerating their exponentially larger ambient word spaces.

The canonical run makes 618,419 exact assertions over 25,384 source transitions. Its final status is PASS. This is regression and counterexample pressure only, not a proof, a process-separated review, an exhaustive owner search, or evidence of novelty. In particular, all classical representation-theoretic identities used in Sections 2–3 remain fully subtracted, as does the crystal pop-stack dynamical surface. The claim ceiling is the finite dynamical conjunction of the literal scheduler (1), the clock (6), and the targetwise inverse rule (13). The package remains OWNER_AMBER / HOLD_EXTERNAL.

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